Unity XR Development

Workshop 2

Workshop 1 Recap & Multimodal Instructions in XR

XR2IND-VR | Unity 2022.3 | Meta Quest 3

Workshop 1 — What We Built

The XR2IND-VR project - a VR training application designed to teach how to install,configure and troubleshoot industrial networking equipment.

XR Project Setup

- Unity 2022 + XR Interaction Toolkit
- Meta Quest 3 deployment via SideQuest
- XR Device Simulator for desktop testing

XR Interactions

- Grab & release objects (XRGrabInteractable)
- Trigger button input bindings

XR2IND Scene

- Juniper & Cisco equipment models
- ScenarioObjectives task tracking system

Workshop 1 gave you the skeleton of a VR training app.

→ **Today: Multimodal Instructions in XR**

Text · Video · Audio · Spatial Audio · Haptics

Workshop - 2 Multimodal Instructions

01	**Text in XR**	World-space Canvas · TextMeshPro · driving text from script
02	**Video**	VideoPlayer · RenderTexture · streaming from URL/StreamingAssets
03	**Audio & Spatial**	AudioSource · Spatial Blend 0 vs 1 · Min/Max Distance rolloff
04	**Haptics**	Single Impulse · Continuous · Pattern - 3 cubes, 3 types

Workshop 2 — Multimodal Instructions

Text in XR

Two ways to show the text :

- it can live on a canvas (UI space) or as a textmeshpro object in world space

Video

Unity's Video Player component renders video onto a Render Texture, which is then displayed on a material. This means you can play video on any surface in your 3D world: a screen on the wall, a panel floating in VR, or a flat surface on a desk.

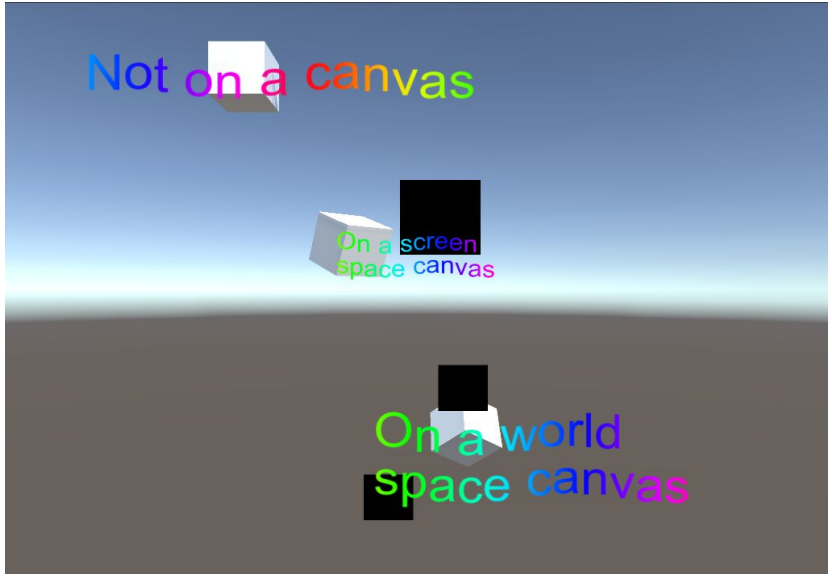
Audio

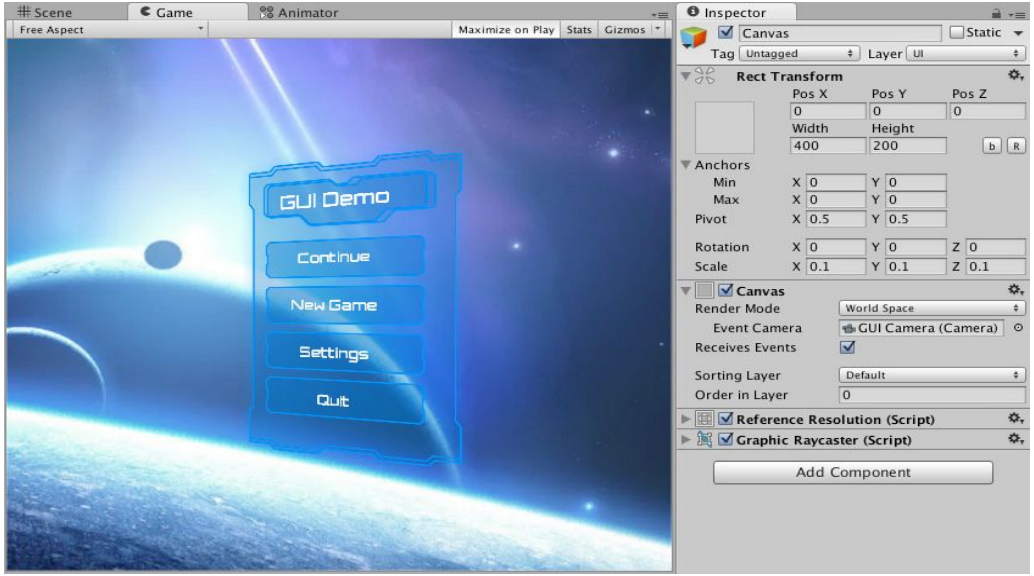
In the real world, sounds come from two places: they either follow you everywhere (like music in headphones) or they come from a specific location in space . In VR we replicate both. Unity calls these 2D audio and 3D spatial audio.

Haptics

Haptics are vibrotactile feedback delivered through the controller to the user's hand. They are one of the most powerful presence cues in VR

Text in XR


Not on a canvas
On a screen space canvas
On a world space canvas


Scene
Game
Animator
Free Aspect
Maximize on Play
Stats
Gizmos
GUI Demo
Continue
New Game
Settings
Quit
Inspector
Canvas
Static
Tag Untagged
Layer UI
Rect Transform
Pos X 0
Pos Y 0
Pos Z 0
Width 400
Height 200
Anchors
Min X 0 Y 0
Max X 0 Y 0
Pivot X 0.5 Y 0.5
Rotation X 0 Y 0 Z 0
Scale X 0.1 Y 0.1 Z 0.1
Canvas
Render Mode World Space
Event Camera GUI Camera (Camera)
Receives Events
Sorting Layer Default
Order in Layer 0
Reference Resolution (Script)
Graphic Raycaster (Script)
Add Component

Three types of haptic feedback in XR

Type	Description	Real-world analogy
Single Impulse	One short buzz at set amplitude & duration	Clicking a physical button, instant click sensation
Continuous	Sustained vibration while holding/touching	Holding a running electric drill , constant hum
Pattern / Pulse	Sequence of on/off bursts with timing control	Morse code or phone notification rhythm

Other haptics which can be used

A) Intensity ramp (fade in/out)

Like a motor spinning up. gradually change amplitude over time.

```
IEnumerator Ramp(ActionBasedController c, float start, float end, float seconds)
{
    float t = 0;
    while (t < seconds)
    {
        float a = Mathf.Lerp(start, end, t / seconds);
        c.SendHapticImpulse(a, 0.02f);
        t += 0.02f;
        yield return new WaitForSeconds(0.02f);
    }
}
```

Use for: drill speeding up, tension building, charging.

B) Distance-based haptics (hot/cold proximity)

Stronger vibration as you get closer to something. amplitude = function(distance)

```
void Update()
{
    if (heldBy == null) return;

    float d = Vector3.Distance(transform.position, target.position);
    float a = Mathf.Clamp01(1f - d / 0.5f); // within 0.5m
    heldBy.SendHapticImpulse(a, 0.02f);
}
```

Use for: guiding users to alignment points, “magnetic” snapping cues.

C) Impact haptics (collision strength)

Harder collision = stronger pulse. use collision relative velocity

```
void OnCollisionEnter(Collision col)
{
    var c = heldBy; if (c == null) return;
    float strength = Mathf.Clamp01(col.relativeVelocity.magnitude / 3f);
    c.SendHapticImpulse(strength, 0.08f);
}
```

Use for: tool hits surface, dropping object, bump feedback.

D) “Texture” haptics (scraping / bumps)

Feels like moving across rough surface. small pulses at a frequency while sliding.

```
IEnumerator TextureBuzz(ActionBasedController c)
{
    while (true)
    {
        c.SendHapticImpulse(0.15f, 0.01f);
        yield return new WaitForSeconds(0.03f); // frequency
    }
}
```

Use for: sanding, scraping, sawing.

E) Two-hand / two-controller haptics

If both hands touch/hold, vibrate both.Track both controllers and send to each.